SAP SECURITY ADMINISTRATION, OPERATIONS & TROUBLESHOOTING

Chapter 3 – RFC Security, Background Jobs, Interface Users & Technical Security

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Q216. What is RFC?

Interviewer's Objective

The interviewer wants to assess whether you understand RFC as a secure communication mechanism rather than just a Basis configuration.

Executive Answer

RFC (Remote Function Call) is SAP's standard communication protocol that enables one SAP system or an external application to execute functions in another SAP system.

RFC is used for:

- **System-to-system communication**
- **SAP GRC integration**
- **SAP IDM provisioning**
- **ALE/IDoc processing**
- **Background jobs**
- **External interfaces**
- **Middleware integration**

Since RFC users often have privileged access, securing RFC connections is a critical SAP Security responsibility.

Typical Architecture

```
SAP GRC
  │
  ▼
RFC Connection
  │
  ▼
SAP S/4HANA
```

Q217. Explain the different RFC Types.

Executive Answer

SAP supports several RFC connection types.

RFC Type	Purpose
Type 3	ABAP Connection (Most Common)
Type G	HTTP Connection
Type T	TCP/IP Connection
Type H	HTTP via SAP Router
Type L	Logical Destination

Interview Tip

For SAP Security and GRC interviews, Type 3 (ABAP Connection) is the most important because it is used extensively for GRC connectors, IDM, and inter-system communication.

Q218. What is SM59?

Very common interview question.

Executive Answer

SM59 is the SAP transaction used to create, configure, and test RFC destinations.

Using SM59, administrators can:

- **Create RFC destinations**
- **Test connectivity**
- **Configure logon credentials**
- **Configure trusted RFCs**
- **Enable Secure Network Communication (SNC)**

- **Maintain timeout settings**

Common Tests

- **Connection Test**
- **Authorization Test**
- **Unicode Test**

Q219. What is a Trusted RFC?

Executive Answer

A Trusted RFC allows one SAP system to trust another SAP system without requiring the user to enter credentials repeatedly.

Example:

SAP GRC

↓

Trusted RFC

↓

SAP S/4HANA

Benefits:

- **Improved user experience**
- **Single Sign-On**
- **Reduced password management**

However, Trusted RFCs must be tightly controlled because compromise of one trusted system may affect connected systems.

Q220. What is Secure Network Communication (SNC)?

Architect-level question.

Executive Answer

SNC (Secure Network Communication) encrypts communication between SAP systems and clients.

It provides:

- **Authentication**
- **Encryption**
- **Integrity protection**
- **Single Sign-On support**

Without SNC:

RFC traffic may be transmitted in clear text, increasing security risk.

Business Benefits

- **Encrypted communication**
- **Compliance with security standards**
- **Protection against network attacks**

Q221. How do you secure RFC Connections?

Executive Answer

I recommend a layered approach:

1. **Use dedicated technical users.**
2. **Apply the Principle of Least Privilege.**
3. **Enable SNC wherever possible.**
4. **Avoid hard-coded passwords.**
5. **Restrict trusted RFC relationships.**
6. **Monitor RFC activity regularly.**
7. **Review unused RFC destinations periodically.**

8. **Protect RFC users with strong password policies where applicable.**

Q222. Explain Background Job Security.

Executive Answer

Background jobs execute using the authorizations of the scheduling user or the technical user under which they run.

Security considerations include:

- **Dedicated batch users**
- **Least privilege**
- **Secure variants**
- **Job ownership**
- **Monitoring**
- **Periodic review**

Key Transactions

- **SM36 – Schedule Jobs**
- **SM37 – Monitor Jobs**

Q223. What are Technical Users?

Executive Answer

Technical Users are non-human accounts used for system communication and automated processing.

Examples:

- **RFC users**
- **Batch users**
- **Interface users**
- **Service accounts**

- **Middleware users**

These accounts should never be used for interactive business activities.

Q224. Explain Batch Users.

Executive Answer

Batch users are designed to execute scheduled jobs.

Characteristics:

- **No interactive logon**
- **Used by background processing**
- **Dedicated authorizations**
- **Restricted access**

Each batch user should be assigned only the authorizations required for the jobs it executes.

Q225. Explain Interface User Security.

Executive Answer

Interface users facilitate communication between SAP and external systems.

Examples:

- **SAP PI/PO**
- **SAP CPI**
- **Middleware**
- **Third-party applications**

Security Best Practices:

- **Dedicated accounts**
- **Least privilege**
- **Password rotation**

- Monitoring
- Logging
- Periodic review

Q226. Explain SAP Router.

Executive Answer

SAP Router is a secure gateway that controls network communication between SAP systems.

Functions include:

- Network routing
- Access control
- Connection filtering
- Firewall-like protection

SAP Router reduces direct exposure of SAP systems to external networks.

Q227. Explain Gateway Security.

Executive Answer

The SAP Gateway manages RFC communication and external program registration.

Gateway security focuses on:

- Restricting registered external programs
- Limiting trusted hosts
- Preventing unauthorized RFC registrations
- Monitoring gateway logs

Misconfigured gateways can expose systems to unauthorized remote execution.

Q228. Explain RFC User Governance.

Executive Answer

RFC users should be governed like privileged accounts.

Key controls include:

- **Business owner assignment**
- **Purpose documentation**
- **Periodic access review**
- **Password management**
- **Monitoring**
- **Removal of unused accounts**
- **Audit logging**

Q229. Production Scenario

Interviewer

"SAP GRC suddenly cannot connect to the Production S/4HANA system. What would you check?"

Executive Answer

I would investigate in the following order:

1. **Test the RFC destination in SM59.**
2. **Review network connectivity.**
3. **Verify the RFC user's status and password validity.**
4. **Confirm the target system is available.**
5. **Check gateway and work process logs.**
6. **Validate SNC configuration if enabled.**
7. **Review recent transports or Basis changes affecting RFC destinations.**
8. **Confirm authorizations for the RFC user.**

This structured approach isolates connectivity, authentication, authorization, and infrastructure issues.

Q230. Real Manufacturing Scenario

Interviewer

"A nightly interface that transfers production orders from SAP S/4HANA to MES suddenly fails. Business reports no production updates. How would you troubleshoot?"

Executive Answer

I would approach this as both a business-critical and technical incident.

1. **Determine whether the failure is due to the interface, RFC communication, or application processing.**
2. **Check SM37 for failed background jobs.**
3. **Review SM59 connectivity to the MES interface.**
4. **Validate the interface user's status and authorizations.**
5. **Review gateway logs and middleware logs (SAP PI/PO, CPI, or other integration platform if applicable).**
6. **Verify that no recent transports affected RFC destinations or interface authorizations.**
7. **Work with the manufacturing functional team to determine whether business data or configuration changes contributed to the failure.**
8. **Restore the interface, validate successful data transfer, and document the root cause.**

Rapid-Fire Questions

Which transaction maintains RFC destinations?

Answer: SM59

Which transaction schedules background jobs?

Answer: SM36

Which transaction monitors background jobs?

Answer: SM37

What protocol does SAP use for Remote Function Calls?

Answer: RFC (Remote Function Call)

What encrypts SAP communication?

Answer: SNC (Secure Network Communication)

Should RFC users be Dialog users?

Answer: Generally no. Dedicated technical user types should be used according to the integration requirement, with only the minimum authorizations necessary.

Executive-Level Interview Question

Interviewer

"What is the biggest security risk related to RFC connections?"

Executive Answer

The greatest risk is granting excessive privileges to RFC technical users.

Because RFC users often execute privileged operations automatically and are not interactive users, excessive authorizations can remain unnoticed for long periods.

My approach is to:

1. **Assign dedicated RFC accounts.**
2. **Apply the Principle of Least Privilege.**
3. **Enable encrypted communication using SNC where possible.**
4. **Monitor RFC activity and review logs.**
5. **Conduct periodic access reviews.**
6. **Remove unused RFC destinations and dormant technical accounts.**

Strong governance around RFC users significantly reduces the attack surface of the SAP landscape.

Architect's Insight

One of the strongest statements you can make in an interview is:

"Technical users should be treated as privileged identities. Although they are non-human accounts, they often have broader system access than business users and therefore require the same—or greater—level of governance, monitoring, and periodic review."

That demonstrates an enterprise security mindset rather than just SAP administration knowledge.

Chapter Summary

By the end of this chapter, you should be comfortable explaining:

- **RFC architecture**
- **RFC types**
- **SM59**
- **Trusted RFC**
- **SNC**
- **Background job security**
- **Technical users**
- **Batch users**
- **Interface users**
- **SAP Router**
- **Gateway security**
- **RFC governance**
- **Production troubleshooting**
- **Manufacturing integration scenarios**

Next Chapter – SAP Audit, Compliance, Logging & Monitoring (SM19, SM20, STAD, Security Audit Log)

This chapter is another favorite for senior interviews and will cover:

- **Security Audit Log (SM19/SM20)**
- **STAD workload analysis**
- **Change documents**
- **User Information System (SUIM)**
- **Audit evidence collection**
- **SOX & ITGC controls**
- **Compliance reporting**
- **Continuous monitoring**
- **Audit preparation**
- **Real auditor interview scenarios**
- **Production incident investigations**

This is particularly valuable for SAP GRC Security Lead and Architect roles because it demonstrates not only technical knowledge but also audit and compliance expertise.